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KisaanMadad: An Integrated Mobile Decision-Support System for Small-Scale Farmers in Punjab, Pakistan

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Abstract

Small-scale farmers in Pakistan face a confluence of structural challenges that severely constrain agricultural productivity and household income: acute water scarcity, persistent crop price volatility, and the near-total absence of data-driven decision-support tools accessible at the farm level. This report presents KisaanMadad ("Farmer Help"), an integrated mobile decision-support system designed for resource-constrained smallholders in Punjab, Pakistan. The system unifies four functionally distinct modules within a mobile architecture that prioritises a native Android client built with Jetpack Compose for the primary deployment environment, while retaining Kotlin Multiplatform shared logic to support iPhone interoperability: a probabilistic price forecasting engine employing a CNN-LSTM hybrid architecture benchmarked against Prophet and ARIMA baselines; a multi-criteria crop recommendation module that synthesises real-time weather, market proximity, planting calendars, and water availability; an irrigation scheduling engine grounded in the FAO-56 Penman-Monteith method, computing reference evapotranspiration ET_0 and applying crop-specific coefficients K_c to derive field-level water requirements; and a financial planning toolkit comprising loan calculators, revenue projections, and expense tracking. All modules draw on localised data sources — Punjab mandi price records from PAMRA and AMIS alongside meteorological inputs from the NASA POWER API — and are presented through an Urdu-first interface to serve farmers with limited digital literacy. Collectively, KisaanMadad is expected to reduce the information asymmetry experienced by smallholders, improve irrigation efficiency, and provide actionable forecasts that enable more profitable and climate-resilient farming decisions.

Executive Summary

Agriculture underpins Pakistan's economy, contributing 23.54% to gross domestic product and employing 37.4% of the national labour force [1]. Yet the sector operates under severe structural stress. Approximately 97% of Pakistani farmers own fewer than 12.5 acres of land [2], placing them squarely in the smallholder category where exposure to market shocks is highest and access to formal advisory services is lowest. Water insecurity compounds this vulnerability: per capita availability has collapsed from 5,260 m^3 in 1951 to 908 m^3 in 2021 — below the internationally recognised scarcity threshold of 1,000 m^3 [3] — while irrigation efficiency across the canal network remains below 40%. Price volatility driven by macroeconomic instability and strong commodity spillover effects further erodes farm income. Despite these pressures, the existing Pakistani agri-technology landscape remains fragmented: applications address either price discovery or weather information in isolation, and no reviewed solution integrates financial planning tools alongside agronomic advisory functions.

KisaanMadad is an Android-first mobile application targeting wheat, cotton, and rice cultivation in Punjab, Pakistan. It consolidates four decision-support modules into a unified, Urdu-language interface designed for farmers with limited digital literacy. The primary client is implemented natively with Jetpack Compose so that the project can be tuned for the low-cost Android handsets most likely to be encountered in field deployment, while Kotlin Multiplatform is adopted for shared domain and networking logic so that an iOS client can reuse the same core services with limited duplication [4, 5, 6]. The Price Forecasting module delivers probabilistic commodity price predictions for major Punjab mandi markets. The Crop Recommendation module generates ranked planting suggestions by scoring candidate crops against current weather conditions, market access, seasonal calendars, and water availability. The Irrigation Scheduling module produces weekly irrigation schedules tailored to a farmer's location, crop growth stage, and prevailing meteorological conditions. Finally, the Financial Planning module provides loan repayment calculators, crop revenue projections, and seasonal expense trackers — functionality absent from all competing applications reviewed during this study.

The price forecasting engine is built on a CNN-LSTM hybrid architecture, which exploits convolutional layers for local temporal pattern extraction and long short-term memory layers for longer-range dependency modelling. The model is trained on daily wholesale price records sourced from PAMRA and AMIS spanning 2018–2023 and benchmarked against Prophet and ARIMA baselines [7]. Irrigation scheduling is grounded in the FAO-56 Penman-Monteith standard [8], computing reference evapotranspiration ET_0 from NASA POWER API solar radiation, temperature, and precipitation data, then multiplying by the appropriate crop coefficient K_c to yield crop evapotranspiration $ET_c = K_c \times ET_0$; a Hargreaves–Samani fallback is applied when complete radiation data are unavailable. Crop recom-

mendation scoring combines these meteorological inputs with FAO crop calendar data and proximity weighting for nearby mandis. The financial module employs rule-based calculators parameterised from user-provided inputs and prevailing crop revenue models. At the application layer, the presentation tier is implemented with Jetpack Compose on Android, while Kotlin Multiplatform shares non-UI modules such as API contracts, validation logic, and caching policies with a future iPhone client. This pairing preserves native Android rendering and platform-specific tuning for low-memory devices, yet still offers substantial cross-platform code reuse where business logic is stable [9, 5, 6].

KisaanMadad is expected to provide smallholder farmers with actionable, localised intelligence that reduces reliance on informal intermediaries (arthi) and mitigates the information asymmetry that currently characterises the mandi system. By embedding scientifically grounded irrigation scheduling directly on a mobile handset, the system aims to meaningfully improve on-farm water-use efficiency in a country facing acute water scarcity. Five principal contributions distinguish this work from the existing literature and commercial landscape: (1) it constitutes the first integrated platform to combine price forecasting, crop recommendation, irrigation scheduling, and financial planning within a single Pakistani agri-application; (2) it delivers FAO-56 compliant ET_0 computation on a mobile device for resource-constrained farmers with limited connectivity; (3) it introduces financial planning tools demonstrably absent from all reviewed existing solutions; (4) all models are trained and evaluated on localised PAMRA/AMIS mandi data and NASA POWER meteorological feeds rather than global commodity indices; and (5) the application is built with an Urdu-first interface, prioritising accessibility for the low-literacy smallholder population it serves.

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Chapter 1 Introduction

Agriculture constitutes the backbone of Pakistan's economy, contributing 23.54% to the national Gross Domestic Product (GDP) and employing 37.4% of the country's total labour force [1]. Despite this critical macroeconomic role, the sector is predominantly reliant on smallholder farmers, with 97% of farmers owning less than 12.5 acres of land [2]. These resource-constrained agriculturalists face profound systemic challenges that severely limit their productivity and profitability. Consequently, there remains a substantial yield gap of 40–60% for major crops, such as wheat and sugarcane, when compared to the potential yields achieved in controlled research trials [10].

A primary driver of this agricultural underperformance is the intensifying water crisis. Pakistan's per capita water availability has precipitously declined from 5,260 m^3 in 1951 to a critically low 908 m^3 in 2021, falling below the internationally recognised water scarcity threshold of 1,000 m^3 [3]. Compounding this scarcity is an acute lack of water management; overall irrigation efficiency remains below 40%, and canal conveyance losses are estimated to reach approximately 38% [3]. Furthermore, farmers' livelihoods are continuously threatened by extreme price volatility in local wholesale agricultural markets (*mandis*), a phenomenon deeply exacerbated by macroeconomic instability and strong spillover effects between core commodities like wheat and rice [11].

Although several agricultural technology solutions exist in the Pakistani market, they remain highly fragmented. Existing applications typically offer either weather forecasts or market prices in isolation, leaving a void for an integrated decision-support platform [12]. Most critically, none of the reviewed existing mobile solutions provide localised financial planning tools for smallholder farmers [12]. To address these pressing systemic issues, KisaanMadad is proposed as an integrated mobile decision-support system tailored for small-scale Pakistani farmers, aiming to bridge the information gap and promote sustainable, data-driven agricultural practices.

1.1 Purpose of this Document

The purpose of this document is to present the comprehensive design, implementation, and evaluation of KisaanMadad. In the face of fragmented existing solutions, KisaanMadad consolidates multiple disparate agricultural data streams into a unified, user-centric platform tailored for the specific needs of resource-constrained farmers in Punjab, Pakistan. Thus, the primary research question addressed by this project is whether an integrated, mobile-based decision-support system can effectively deliver localised crop recommendations, precise irrigation scheduling, accurate market price forecasting, and crucial financial planning directly to smallholder farmers. To answer this, the document details the development

of the system's four core modules: ML-based Price Forecasting, multi-criteria Crop Recommendation, FAO-56 compliant Irrigation Scheduling, and rule-based Financial Planning.

1.2 Intended Audience

The primary intended audience for this document is the evaluation committee at the National University of Computer and Emerging Sciences (FAST-NUCES), Lahore, who will assess the technical rigour and academic contribution of the KisaanMadad Final Year Project. Additionally, the audience encompasses potential agricultural stakeholders, including government bodies like the Punjab Agriculture Marketing Regulatory Authority (PAMRA), non-governmental organisations, and agri-tech innovators. These entities will gain valuable insights into how integrating data-driven mobile technologies can empower small-scale farmers, optimise resource allocation, and enhance agricultural economic stability in Pakistan.

1.3 Definitions, Acronyms, and Abbreviations

Definitions of key terms, along with the expansions of acronyms and abbreviations used throughout this report, are provided below.

1.3.1 Definitions

Mandi A local wholesale agricultural market where crop trading and price determination occur.

Arthi A middleman or commission agent operating within the mandi system, often facilitating transactions between farmers and buyers.

ET_0 Reference evapotranspiration, representing the environmental evaporative demand for a standard reference crop.

K_c Crop coefficient, a standard value representing the physiological characteristics of a specific crop used to determine crop evapotranspiration.

FAO-56 The standard Penman-Monteith method for computing crop water requirements as outlined by the Food and Agriculture Organization [8].

DSS Decision-Support System, an information system that supports business or organisational decision-making activities.

Smallholder farmer An agriculturalist owning or managing a small plot of land, typically less than 12.5 acres in the context of Pakistan, often constrained by limited resources.

1.3.2 Acronyms and Abbreviations

PAMRA Punjab Agriculture Marketing Regulatory Authority

AMIS Agriculture Marketing Information Service

NASA POWER NASA Prediction of Worldwide Energy Resources API

CNN-LSTM Convolutional Neural Network and Long Short-Term Memory hybrid

FAO Food and Agriculture Organization of the United Nations

DSS Decision-Support System

FYP Final Year Project

SDG Sustainable Development Goal

1.4 Conclusion

To conclude, this chapter has provided a foundational understanding of the profound systemic challenges impeding Pakistan's agriculture sector, including severe water scarcity, widespread yield gaps, and significant market price volatility. The fragmented nature of current digital interventions necessitates the development of KisaanMadad, an integrated mobile decision-support system aimed at empowering smallholder farmers. The remaining document is structured to provide a comprehensive exploration of this proposed solution. Chapter 2 delves into an extensive literature review and analysis of related work. Chapter 3 discusses the system design and architecture of the proposed platform. Chapter 4 outlines the specific methodology employed for each of the four core modules. Chapter 5 details the technical implementation of the system. Chapter 6 presents the results and evaluation of the developed modules. Finally, Chapter 7 provides the conclusion and highlights avenues for future work.

Chapter 2 Project Vision

2.1 Problem Domain Overview

Agriculture constitutes 23.54% of Pakistan's GDP and employs over a third of its workforce [1]. However, an estimated 97% of all farmers in Pakistan own and cultivate less than 12.5 acres of land [2], dominating the sector with smallholder operations that lack access to modern tools and data-driven insights. This productivity deficit results in severe yield gaps of 40-60% [13]. Our integrated mobile application centralises essential agricultural decision-making processes—from crop selection and irrigation scheduling to price forecasting and financial planning. By deploying these modules collectively, the system ensures more reliable, timely, and actionable intelligence for farmers, ultimately enhancing both individual profitability and broader regional food security.

2.2 Problem Statement

The problem at hand involves the persistent reliance of Pakistani smallholder farmers on fragmented, traditional heuristics rather than integrated, data-driven tools. This deficiency results in suboptimal agricultural decisions across crop rotation, water utilisation, and market timing. By developing a unified, Urdu-first mobile decision-support system, we aim to automate complex predictive models and deliver actionable insights, thereby mitigating risks and improving overall farm efficiency.

2.3 Problem Elaboration

The overarching problem is characterised by several critical sub-domains that severely limit agricultural productivity and profitability.

2.3.1 Market Information Asymmetry

Smallholders face extreme price volatility and lack direct access to forward-looking, real-time wholesale mandi prices. Existing applications fail to provide predictive insights using advanced machine learning models (e.g., CNN-LSTM) [14], leaving farmers vulnerable to exploitation by intermediaries [11] and unable to optimally time their sales.

2.3.2 Irrigation Inefficiency

Pakistan faces a severe hydrological crisis [3], yet agricultural water use remains highly inefficient due to a reliance on traditional flood irrigation that causes massive volumetric waste [15]. There is a distinct

absence of accessible mobile tools that utilise scientifically validated evapotranspiration models, such as the FAO-56 standard [8], to guide precise and efficient watering schedules.

2.3.3 Fragmented Tools

Current digital agricultural interventions are heavily siloed, offering discrete services like basic weather or isolated tips rather than a holistic platform [12]. Furthermore, these tools often feature complex English interfaces, making them largely inaccessible to the vast majority of rural farmers who require intuitive, Urdu-first design to overcome digital literacy barriers [16].

2.3.4 Financial Exclusion

Smallholder farmers suffer from pervasive financial exclusion and a lack of formal planning capacity [17]. The conspicuous absence of comprehensive financial modules in existing agri-tech applications heavily forces farmers to make critical livelihood decisions based purely on intuition rather than mathematically projected revenues, tracking inputs, and managing expenses.

2.4 Goals and Objectives

The goals and objectives of this project, addressing the lack of integrated decision-support for farmers, are as follows:

1. Develop a probabilistic machine learning pipeline, benchmarking CNN-LSTM against baseline models [7], to accurately forecast mandi crop prices.
2. Implement a multi-criteria scoring algorithm for automated, localized crop recommendation.
3. Design an irrigation scheduling module based on the FAO-56 Penman-Monteith standard for optimal water management.
4. Engineer a comprehensive financial planning module for revenue projection and expense tracking.
5. Deploy the system as an accessible, native Urdu-first mobile application.

2.5 Project Scope

The project encompasses two distinct phases: the research phase and the development phase. The research phase involves a comprehensive literature review to gather insights on existing agricultural DSS, predictive modeling, and water management strategies. Additionally, it involves identifying reliable

open-source data repositories, such as AMIS, PAMRA, and NASA POWER, that will serve as foundational inputs for our system.

The development phase focuses on creating the KisaanMadad platform, scoped geographically to Punjab, Pakistan, for wheat, cotton, and rice. The software encompasses four integrated modules: ML-based price forecasting, multi-criteria crop recommendation, FAO-56 compliant irrigation scheduling, and offline-capable financial calculators. The final deliverable is an Urdu-supported mobile application. Real-time physical IoT sensors, computer vision disease detection, and direct micro-lending integrations are strictly excluded from this scope.

2.6 Sustainable Development Goals

In this project, we are focusing on two specific SDGs: Zero Hunger, and Industry, Innovation, and Infrastructure.

2.6.1 SDG 2: Zero Hunger

Figure 2.1: United Nations Sustainable Development Goal 2: Zero Hunger.

The first SDG is Zero Hunger. By actively providing vulnerable rural farmers with transparent data-driven crop recommendations and sophisticated forward-looking price forecasts, the integrated system directly addresses foundational knowledge gaps. This project supports the goal of doubling the agricultural productivity and baseline incomes of small-scale food producers [18], promoting resilient practices that adapt to climate change.

Figure 2.2: United Nations Sustainable Development Goal 9: Industry, Innovation, and Infrastructure.

2.6.2 SDG 9: Industry, Innovation, and Infrastructure

The second SDG we address is Industry, Innovation, and Infrastructure. Our solution leverages innovative computational models—like CNN-LSTM forecasts and FAO-56 calculations [8]—and translates them into a highly accessible mobile interface. By doing so, we contribute to upgrading the fundamental technological capabilities of the agricultural sector, bridging the digital divide for rural workers without demanding prohibitive infrastructure investments [18].

2.7 Conclusion

To conclude, this chapter establishes the foundational project vision by contextualising the profound challenges faced by smallholder farmers in Pakistan, including market volatility, water scarcity, and fragmented tools. By clearly articulating an unambiguous problem statement, identifying highly specific objectives, and formally defining the project scope, the chapter demonstrates how an integrated decision-support system can simultaneously advance critical UN Sustainable Development Goals and empower end-users. This sets the stage for a comprehensive exploration of our multifaceted approach in subsequent chapters.

Chapter 3 Literature Review / Related Work

3.1 Definitions, Acronyms and Abbreviations

3.1.1 Definitions

Evapotranspiration The combined process of water loss from the earth's surface into the atmosphere through evaporation from soil and transpiration from plants.

Reference Evapotranspiration (ET_0) The evapotranspiration rate from a well-watered reference surface, typically an extensive surface of green grass of uniform height.

Crop Coefficient (K_c) A dimensionless ratio that relates the reference evapotranspiration to the actual crop evapotranspiration under standard conditions.

Mandi A local or regional wholesale agricultural market where farmers sell their produce to commission agents or buyers.

Time Series Forecasting The use of mathematical and statistical models to predict future values based on previously observed, chronologically ordered data.

Multi-Criteria Decision Making (MCDM) A sub-discipline of operations research that explicitly evaluates multiple, often conflicting, criteria in decision making.

Soil Moisture The quantity of water contained in the soil matrix, which is a critical variable for determining irrigation triggers.

Precision Agriculture A farming management concept based on observing, measuring, and responding to inter- and intra-field variability in crops.

Vernacular Interface A user interface localized into a native or indigenous language, tailored directly to users who do not speak English.

3.1.2 Acronyms and Abbreviations

ANN Artificial Neural Network

ARIMA AutoRegressive Integrated Moving Average

CNN Convolutional Neural Network

DSS Decision Support System

ET_0 Reference Evapotranspiration

FAO Food and Agriculture Organization

GIS Geographic Information System

LSTM Long Short-Term Memory

MAE Mean Absolute Error

MAPE Mean Absolute Percentage Error

MCDM Multi-Criteria Decision Making

ML Machine Learning

NASA POWER NASA Prediction of Worldwide Energy Resources

PAMRA Punjab Agriculture Marketing Regulatory Authority

RMSE Root Mean Square Error

UI User Interface

UX User Experience

3.2 Detailed Literature Review

3.2.1 Agricultural Price Forecasting

3.2.1.1 Summary of the research items

Menculini et al. [7] propose comparing deep learning methods such as Prophet alongside traditional ARIMA architectures for forecasting wholesale food prices. Using a large dataset of regional wholesale market prices, the study demonstrates that deep learning significantly outperforms classical statistical techniques, achieving lower error rates across different food commodity models.

In [19], Murugesan and Mishra evaluate various deep learning architectures, including basic LSTM, bi-LSTM, stacked LSTM, and CNN-LSTM configurations, for predicting agricultural commodity prices. By extracting temporal sequences and spatial patterns, this study demonstrates that CNN-LSTM hybrids offer a highly accurate forecasting mechanism, outperforming basic recurrent neural networks in complex market scenarios.

The study by Joshi and Patel [20] utilized a CNN-Bidirectional LSTM approach specifically applied to price forecasting of volatile agricultural commodities in Gujarat. The bidirectional model effectively learned complex non-linear trading patterns, confirming the suitability of deep sequential models to capture both forward and backward temporal dependencies in local agriculture markets.

Finally, Habib et al. [11] analyse the impact of price volatility on agricultural commodities versus general food items with a specific focus on Pakistan. Utilizing localized economic datasets alongside historical prices, the authors demonstrate how macroeconomic instability and international commodity trends compound short-term price fluctuations. Their baseline findings underscore the necessity for robust, localized predictive tools to mitigate these shocks.

3.2.1.2 Critical analysis of the research items

The comparative study by Menculini et al. [7] successfully establishes the supremacy of deep learning over ARIMA; however, its reliance on stable European datasets limits its direct generalisability to highly erratic markets like Pakistan's mandi system. The deep learning models evaluated by Murugesan and Mishra [19] are notably powerful, yet complex architectures such as CNN-LSTM run the risk of overfitting when applied to data-scarce environments. The bidirectional approach from Joshi and Patel [20] handles long-term dependencies gracefully but lacks robust handling of exogenous variables, such as sudden policy shifts or extreme weather shocks, which regularly distort agricultural pricing. Meanwhile, Habib et al. [11] correctly identify structural volatility issues in Pakistan's agriculture sector, but the work remains heavily diagnostic rather than predictive, offering no implementable machine learning framework for regional stakeholders.

3.2.1.3 Relationship to the proposed research work

The CNN-LSTM architecture evaluated positively by Murugesan and Mishra [19] translates to the baseline design for KisaanMadad's price forecasting module, effectively balancing spatial feature extraction with time-series memory similar to the approach by Joshi and Patel [20]. To address the localized volatility highlighted by Habib et al. [11], the module will rely on PAMRA and AMIS daily mandi price datasets spanning 2018–2023, rather than abstract global commodity curves. This bridges the gap between state-of-the-art deep learning findings and the resource-constrained needs of smallholder farmers in Pakistan's localized supply chains.

3.2.1.4 Literature review summary table

3.2.2 Crop Recommendation Systems

3.2.2.1 Summary of the research items

Raghavendra et al. [21] present an extensive systematic literature review of machine learning algorithms utilized strictly for crop recommendation and yield estimation. Evaluating a consolidated body of work, the study demonstrates that ensemble methods and deep learning models consistently outperform tradi-

Table 3.1: Literature review summary table for Agricultural Price Forecasting

Authors	Method	Results	Limitations
Murugesan & Mishra [19]	CNN-LSTM and LSTM variants	Superiority of CNN-LSTM architectures	High computational cost, data heavy
Joshi & Patel [20]	CNN-Bidirectional LSTM	Accurate regional forecasting in Gujarat	Ignores exogenous market shocks
Menculini et al. [7]	Prophet, Deep, ARIMA	Prophet and DL outperformed ARIMA	Tested on stable European datasets
Habib et al. [11]	Economic Analysis	Identified severe structural price volatility	Diagnostic only, no predictive application

tional ML approaches when trained on diverse soil and atmospheric inputs.

In [22], Priyanka et al. present an IoT-based crop recommendation framework using a machine learning backend for smart agriculture. Using sensory data measuring temperature and moisture, this study maps real-time environmental data to crop viability, demonstrating early success in automating recommendations dynamically.

Kancharagunta et al. [23] survey crop suitability systems by combining diverse IoT inputs with machine learning models. The survey highlights robust methodologies that integrate continuous environmental parameters into algorithmic decisions, successfully aiding both yield prediction and agricultural scheduling.

Finally, Apat et al. [24] focuses specifically on artificial intelligence-based frameworks for crop guidance, proposing a lightweight intelligent recommendation system. By classifying secondary datasets encompassing soil and water availability features, the proposed system proves highly capable for farmers lacking expensive localized hardware.

3.2.2.2 Critical analysis of the research items

While Raghavendra et al. [21] provides an excellent foundational synthesis of machine learning in agriculture, the aggregated literature flattens regional heterogeneities, making perfect accuracy difficult to replicate in real-world scenarios. The IoT-driven architecture by Priyanka et al. [22] succeeds functionally but remains entirely cost-prohibitive for the mass majority of farmers owning under 12.5 acres. The survey approach by Kancharagunta et al. [23] confirms these hardware-adoption barriers. Conversely, the methodology by Apat et al. [24] is a contextually robust software-only approach, yet its reliance on secondary static traces typically limits its predictive power against volatile precipitation cycles and climate changes.

3.2.2.3 Relationship to the proposed research work

KisaanMadad's crop recommendation system builds upon the software-centric philosophy presented by Apat et al. [24] while aiming for the robust accuracy highlighted by Raghavendra et al. [21]. To overcome the lack of IoT hardware noted as a vulnerability by Priyanka et al. [22] and Kancharagunta et al. [23], KisaanMadad will integrate dynamic, real-time meteorological inputs via the NASA POWER API alongside FAO crop calendars. By weighing continuous remote variables against static local criteria, KisaanMadad proposes an accurate recommendation system directly tailored to small-scale growers in Punjab.

3.2.2.4 Literature review summary table

Table 3.2: Literature review summary table for Crop Recommendation Systems

Authors	Method	Results	Limitations
Priyanka et al. [22]	IoT-based ML classification	High dynamic accuracy	Hardware-dependent, cost-prohibitive for smallholders
Apat et al. [24]	AI modeling on secondary data	Highly viable crop mapping	Relies heavily on static or prior weather traces
Kancharagunta et al. [23]	Survey of ML/IoT systems	Synthesis of precision agritech adoption	Implementation complexity highlighted in survey
Raghavendra et al. [21]	Systematic literature review	Identified optimal ensemble models	Generalised findings ignore regional variances

3.2.3 FAO-56 Irrigation Scheduling and ET_0 Estimation

3.2.3.1 Summary of the research items

The foundational work by Allen et al. [8] establishes the global standard for computing crop water requirements via the FAO-56 Penman-Monteith method. This exhaustive procedural framework uses defined crop coefficients (K_c) and standard meteorological variables to produce the definitive reference evapotranspiration (ET_0) metric used globally.

In [25], Harris proposes an energy-efficient smart irrigation framework combining edge AI with forecast-driven scheduling. By processing atmospheric variables against datasets like FAO and NASA POWER, the system efficiently minimized irrigation errors and operational costs through anticipatory remote sensing.

A study by Baber and Ullah [26] evaluates the temperature-based Hargreaves-Samani equation against the standard FAO-56 Penman-Monteith equation for ET_0 estimation strictly within Pakistan. Using regional observatory data, this study demonstrates that while the PM method holds theoretical supremacy,

the calibrated Hargreaves-Samani model is exceptionally suitable and accurate for data-scarce Pakistani regions.

Lastly, Jaafar et al. [27] proposes AgSAT, a smart irrigation application reliant on cloud-fetched NASA POWER data combined with satellite imagery. By directly calculating daily crop ET and water requirements via remote methodologies, the authors demonstrate robust irrigation scheduling capabilities without physical ground equipment.

3.2.3.2 Critical analysis of the research items

The canonical methodology provided by Allen et al. [8] remains technically unmatched, yet its rigid demand for numerous meteorological variables makes it exceptionally difficult to directly deploy in sensor-less regions. The edge AI approach by Harris [25] brings powerful computation directly to farms but necessitates localized hardware or continuous connectivity. The regional evaluation by Baber and Ullah [26] highlights a crucial compromise: the Hargreaves equation is simpler, but without proper calibration coefficients, farmers risk wasteful over-irrigation. The utilization of NASA POWER datasets by mobile applications like the one developed by Jaafar et al. [27] presents an optimal software pathway, though resolution constraints inevitably restrict effectiveness on extreme micro-topographies.

3.2.3.3 Relationship to the proposed research work

Following the data sourcing successes of Harris [25] and Jaafar et al. [27], KisaanMadad bypasses physical hardware constraints by querying the NASA POWER API for real-time solar radiation and temperature. This remote data populates the core FAO-56 Penman-Monteith engine [8] compiled within the application. When wind speed or relative humidity inputs fail, KisaanMadad falls back to the temperature-driven Hargreaves methodology verified for use in local contexts by Baber and Ullah [26], guaranteeing uninterrupted irrigation scheduling.

3.2.3.4 Literature review summary table

3.2.4 Agricultural Mobile Applications in Developing Countries

3.2.4.1 Summary of the research items

Kamal and Bablu [28] propose a systematic review regarding the widespread adoption of mobile applications by smallholder farmers operating in developing nations. Their analysis highlights how applications empowering farmers can accelerate agricultural development, provided digital payment systems and information delivery align with ground truths.

In [29], Amina et al. specifically evaluate the factors influencing farmers' satisfaction and adoption of

Table 3.3: Literature review summary table for FAO-56 Irrigation Scheduling

Authors	Method	Results	Limitations
Harris [25]	Edge AI and NASA POWER	Minimized energy load and water use	Requires continuous edge/cloud connectivity
Baber & Ullah [26]	Penman-Monteith vs Hargreaves	Verified Hargreaves-Samani suitability	Calibration issues without proper coefficient setup
Jaafar et al. [27]	NASA POWER & Satellite apps	Strong field-scale ET metrics	Satellite resolutions may overlook micro-climates
Allen et al. [8]	FAO-56 foundational standard	Global ET0 benchmark established	Requires extensive meteorological input variables

mobile phone applications for agricultural purposes within Faisalabad, Pakistan. The study found that usability and reliability directly determine satisfaction, pushing young farmers to optimize operations through reliable, well-designed mobile interfaces.

The investigation by Shahzad et al. [30] examines digital agriculture adoption across Punjab, Pakistan, assessing the role of perceived awareness on farmer decisions. Utilizing a large random sample from diverse agroecological zones, the study proves that localized, culturally coherent outreach dramatically enhances platform adoption and overall digital proficiency.

Finally, examining existing Pakistani platforms, the Pasha [12] sector industry report reviews the state of agricultural tech solutions. The findings emphasize the systemic problem of fragmented single-utility tools, illustrating a severe market need for unified applications capable of managing multiple farming workflows simultaneously.

3.2.4.2 Critical analysis of the research items

The review by Kamal and Bablu [28] comprehensively diagnoses the positive macro-impacts but leaves technical implementation details, like offline caching or UI constraints, unaddressed. The regional usability analysis by Amina et al. [29] correctly identifies young farmers as the initial adoption catalyst in Pakistan, but overlooks the usability constraints faced by aging or low-literacy demographics. The Punjab-centric study by Shahzad et al. [30] provides localized adoption data entirely relevant to Kisaan-Madad, yet highlights awareness, rather than purely technical features, as the core issue. Furthermore, the Pasha [12] sector report critically highlights app fragmentation, emphasizing that single-feature platforms fail to retain user interest without a comprehensive ecosystem backend.

3.2.4.3 Relationship to the proposed research work

Addressing the exact usability hurdles documented by Amina et al. [29] and the necessity of targeted local demographic awareness proven by Shahzad et al. [30], KisaanMadad is natively engineered with

a vernacular Urdu User Interface utilizing accessible, heavily iconized navigational paradigms. Responding directly to the crippling fragmentation decried by the Pasha [12] report and following the empowerment guidelines set by Kamal and Bablu [28], KisaanMadad consolidates price forecasting, crop recommendation, irrigation scheduling, and financial planning into one unified platform, representing a holistic, unprecedented counter to the limitations of current rural deployments.

3.2.4.4 Literature review summary table

Table 3.4: Literature review summary table for Agricultural Mobile Applications

Authors	Method	Results	Limitations
Kamal & Bablu [28]	Review of agricultural dev apps	Proven tech empowerment linkages	Diagnoses macro trends without UI technical guidance
Amina et al. [29]	Faisalabad adoption survey	Identified reliability as key driver	Predominantly evaluates younger demographics
Shahzad et al. [30]	Punjab field adoption stats	Strong correlation with local awareness	Awareness-centric, lacks focus on app mechanics
Pasha [12]	Industry sector reporting	Identified severe app fragmentation	Enterprise focus, no underlying predictive models

3.2.5 Related Applications

3.2.5.1 Summary of the research items

The Kisan App [31] is an established digital advisory platform targeting farmers primarily within Punjab, Pakistan. It focuses heavily on static farming methodologies, regional news alerts, and general weather reporting. It boasts a large user base due to widespread government and telco collaboration, effectively serving as an elementary portal for regional agricultural information.

Farmdar [32] is an established agritech business offering a suite of B2B products (such as CropScan and Yield Pro) rather than a single farmer-facing application. It employs high-resolution, multi-spectral remote sensing AI over satellite imagery to monitor crop health at scale. Deployed heavily across sugar mills, corporate farms, and lending institutions throughout Pakistan and globally, it specializes in actionable spatial detection regarding nutrient deficiencies, pest outbreaks, and yield predictions, driven directly by precision GIS analytics.

BaKhabar Kissan [33] is undoubtedly Pakistan's most prominent holistic agricultural advisory platform. Supported by a robust backend call-center infrastructure and an aggressive telecommunications partnership, the application delivers localized weather parameters, pest warnings, and up-to-date regional

market commodity prices. It caters extensively to smallholder farmers seeking daily operational guidance.

3.2.5.2 Critical analysis of the research items

While the Kisan App [31] is accessible, its feature set is profoundly rudimentary; it severely lacks any advanced machine learning integration for intelligent crop recommendations, and it does not possess price forecasting or financial calculators, rendering it purely informational rather than a true decision support system. Farmdar [32] leverages state-of-the-art spatial precision, yet its enterprise focus, B2B product structure, and heavy reliance on extensive land tracts organically alienate the 97% of farmers confined to holdings under 12.5 acres. Conversely, BaKhabar Kissan [33] addresses smallholders effectively through price reporting, but critically acts only as a diagnostic repository; it fails entirely to forecast incoming price volatility, offers no sophisticated FAO-56 integrated localized irrigation scheduling, and completely omits modular financial planning.

3.2.5.3 Relationship to the proposed research work

The gap analysis of existing entities inherently cements KisaanMadad's unique technological positioning. While BaKhabar Kissan [33] provides basic prices, KisaanMadad introduces dynamic models predicting future prices [19] previously unavailable at zero cost. Unlike the enterprise-focused Farmdar products [32], KisaanMadad explicitly targets smallholders by replacing cost-prohibitive remote B2B analytics with multi-criteria recommendations and FAO-56 [8] frameworks operating natively on mobile. Finally, surpassing the passive information structures of the Kisan App [31], KisaanMadad is the first architecture to fuse agronomic tools with foundational financial planning resources, closing the loop between ecological farm management and smallholder macroeconomic survival.

3.2.5.4 Related applications summary table

Table 3.5: Related applications summary table

Application	Features	Relevance to KisaanMadad	Limitations
Kisan App [31]	News, static advice, weather	General template for UI accessibility	No ML, no forecasting, no financial tools
Farmdar [32]	Remote sensing B2B products	Demonstrates success of precision agritech	Expensive, enterprise-focused, not a direct consumer app
BaKhabar Kissan [33]	Daily prices, pest warnings, call centre	Key competitor in price dissemination	Passive reporting only, no FAO-56 irrigation

Chapter 4 Software Requirements Specification

This chapter outlines the Software Requirements Specification (SRS) for the KisaanMadad integrated mobile decision-support system. It presents the core features, formal functional and non-functional requirements, quality attributes, system constraints, use cases, and database design necessary for deployment in a resource-constrained agricultural environment.

4.1 List of Features

The system provides the following primary features:

- Probabilistic mandi price forecasting with confidence intervals
- Multi-criteria crop recommendation engine
- FAO-56 compliant irrigation scheduling with NASA POWER integration
- Loan calculator and revenue projection tools
- Expense tracker
- Urdu language interface
- Offline-capable core functionality
- Push notifications for price alerts

4.2 Functional Requirements

4.2.1 Price Forecasting Module

1. The system **shall** allow the Farmer to choose a primary crop (wheat, cotton, or rice) and generate a 30-day probabilistic mandi price forecast.
2. The system **shall** retrieve historical daily PAMRA and AMIS records to base its forecasting algorithm outputs.
3. The system **shall** display the corresponding upper and lower confidence intervals alongside the median predicted price chart.
4. The system **shall** trigger push notifications if an anomaly or significant 10% short-term price jump is detected for the Farmer's subscribed crop.

4.2.2 Crop Recommendation Module

1. The system **shall** accept explicit input parameters from the Farmer including land size, district, and planting season.
2. The system **shall** process these geographic parameters via the NASA POWER API to retrieve historical and real-time localized weather data.
3. The system **shall** evaluate inputs systematically against established FAO crop calendars and aggregate environmental metrics.
4. The system **shall** calculate and output a multi-criteria compatibility score representing the suitability of the top recommended crops.

4.2.3 Irrigation Scheduling Module

1. The system **shall** implement the standard FAO-56 Penman-Monteith method to derive baseline reference evapotranspiration (ET_0).
2. The system **shall** parse real-time solar radiation, temperature, and precipitation factors directly from the NASA POWER API payload.
3. The system **shall** apply the designated crop coefficient (K_c) dynamically aligned with the Farmer's active crop growth cycle to determine crop evapotranspiration (ET_c).
4. The system **shall** generate an actionable irrigation schedule detailing required water volume or watering durations.

4.2.4 Financial Planning Module

1. The system **shall** feature an interactive loan calculator requiring interest rate, principal amount, and total tenure to compute uniform monthly installments.
2. The system **shall** process an expected yield input from the Farmer combined with current mandi trading rates to formulate projected revenue.
3. The system **shall** allow the Farmer to record individual input overheads such as seed, fertilizer, and transport costs into an expense tracker.
4. The system **shall** automatically derive and present a projected gross margin comparing accrued tracking expenses against formulated revenues.

4.2.5 Authentication and User Management

1. The system **shall** allow the Farmer to register a distinct account utilizing their mobile number and standard OTP token confirmation.
2. The system **shall** require the Farmer to populate an initial profile detailing primary farm district and language interface preference.
3. The system **shall** securely store established Farmer profile data locally and synchronize passively to the cloud backend.

4.3 Quality Attributes

Table 4.1: System Quality Attributes

Quality Attribute	Description	Relevance to KisaanMadad
Performance	How rapidly the system resolves functions and loads screens.	Critical for acceptable price forecast loading.
Reliability	Operational continuity free from unexpected errors.	Vital for daily irrigation scheduling reliability.
Usability	Clarity and operational ease of the user interface.	Requires Urdu, icons, and non-complex structural design.
Availability	Consistency of application and backend accessibility.	Necessitates offline resilience in rural Punjab.
Scalability	Capability to expand user bases and data payloads.	Adapts to expanding PAMRA archives and Farmer registrations.
Maintainability	Ease of subsequent application modification and patching.	Allows systematic swapping of evolving inference models.
Portability	Environmental independence for multi-device execution.	Achieved through Kotlin Multiplatform shared domain modules while preserving native platform interfaces.

4.4 Non-Functional Requirements

4.4.1 Performance

- The price forecast query execution **shall** complete and return graphic visuals within 3 seconds maximum on a standard 3G/4G cellular network.
- The application **shall** present core features using offline local cache instantly under scenarios with zero active connectivity.

4.4.2 Usability

- The UI **shall** integrate a fully functional vernacular Urdu framework employing familiar iconography to accommodate low-literacy farmers.
- The application interface **shall** limit navigational depth, ensuring any singular core feature is accessible within a maximum of 3 taps from the Home Dashboard.
- On-screen visual elements and typefaces **shall** remain completely legible on conventional 5-inch low-resolution Android display panels.

4.4.3 Reliability

- The overarching API services layer **shall** maintain an uptime target of 99.9% throughout critical regional harvesting months.
- The system **shall** implement graceful degradation, switching automatically to the Hargreaves protocol utilizing local historical temperatures if NASA POWER API endpoints become unresponsive.

4.4.4 Security

- The system **shall** preserve stringent Farmer data privacy policies, collecting zero arbitrary Personally Identifiable Information (PII) without transparent and explicit digital consent.
- All external and internal backend network transmissions **shall** strictly utilize encrypted TLS implementations by default.

4.5 Assumptions

1. The Farmer possesses internet connectivity initially to accommodate application download and primary agricultural data synchronization.
2. The deployment environment consists of low to mid-range Android smartphones running a minimum OS version of Android 8.0.
3. Continuous and un-throttled availability of the external NASA POWER API data payloads is granted during query cycles.
4. Key government and public sources like PAMRA and AMIS will persistently offer stable historic market pricing for the core commodities in Punjab.

5. The Farmer exhibits a baseline level of smartphone operational literacy required to manipulate touchscreen interfaces.

6. Data pulled via longitude and latitude coordinate boundaries from NASA matches the respective farm's atmospheric microclimate with minimal deviation.

4.6 Use Cases

Table 4.2: Use Case: View Price Forecast

Name	View Price Forecast
Actors	Farmer
Summary	Farmer accesses a projected moving price trend for an agricultural commodity.
Pre-Conditions	Appropriate models are cached; UI is operational.
Post-Conditions	Forecast graph visually highlights predictions and intervals.
Special Requirements	The chart visualizes a clear median trend surrounded by intervals.
Basic Flow	
Actor Action	**System Response**
1. Selects desired crop icon	2. Prompts ML model for predictive matrices 3. Plots timeline chart and upper/lower arrays
Alternative Flow	
2a. If local time-series data is stale, the app pushes an alert guiding the user to connect and refresh the cached history prior to execution.	

Table 4.3: Use Case: Get Crop Recommendation

Name	Get Crop Recommendation
Actors	Farmer
Summary	Input localized metrics to receive ranked crop suggestions.
Pre-Conditions	The device possesses network authority to fetch NASA POWER parameters.
Post-Conditions	Screen enumerates recommended crops by descending score.
Special Requirements	Employs FAO crop calendars internally.
Basic Flow	
Actor Action	**System Response**
1. Inputs land acres and regional district	2. Gathers constraints and invokes scoring mechanism 3. Returns the formulated and ranked list to the display
Alternative Flow	
2a. If explicit geographical location polling fails, the UI requests manual district confirmation before proceeding.	

Table 4.4: Use Case: Generate Irrigation Schedule

Name	Generate Irrigation Schedule
Actors	Farmer
Summary	Determine temporal watering cycles via FAO-56.
Pre-Conditions	Farmer specifies identical active crop phase and planting baseline.
Post-Conditions	System delivers specific watering periods or volume suggestions.
Special Requirements	Real-time precision of evapotranspiration calculations.
Basic Flow	
Actor Action	**System Response**
1. Taps on Irrigation module 3. Confirms operation	2. Validates crop and parses NASA POWER weather objects 4. Resolves Penman-Monteith equation and produces the calendar
Alternative Flow	
4a. Unreachable NASA endpoints systematically divert calculation pathways to utilize local historical extrema variables against the Hargreaves method.	

Table 4.5: Use Case: Calculate Loan / Revenue Projection

Name	Calculate Loan / Revenue Projection
Actors	Farmer
Summary	Assess future revenue potentials or amortized banking loan structures.
Pre-Conditions	Up-to-date mandi prices are downloaded within the local state space.
Post-Conditions	Mathematical integers translate to presented financial layouts.
Special Requirements	The interface reflects numeric integers efficiently and cleanly.
Basic Flow	
Actor Action	**System Response**
1. Accesses Financial tools pane 3. Triggers Revenue slider element	2. Presents dual isolated modes 4. Multiplies expected yield by mandi constants live
Alternative Flow	
3a. If the Farmer interacts with the Loan array, the module requests Principal, Interest rate constraints to formulate the cyclic installment value instantly.	

Table 4.6: Use Case: Set Up Farmer Profile

Name	Set Up Farmer Profile
Actors	Farmer
Summary	Initializing the application state parameters corresponding to user specifics.
Pre-Conditions	The primary OTP handshake sequence has concluded.
Post-Conditions	An active profile is structured within standard DB formats.
Special Requirements	Non-intrusive collection of PII.
Basic Flow	
Actor Action	**System Response**
1. Submits localized preferences	2. Translates object states to persistent localized caching 3. Synchronizes background sequence to corresponding remote records
Alternative Flow	
3a. Application is entirely offline; local states update identically and cloud push cycles are suspended pending affirmative connectivity pulses.	

Table 4.7: Use Case: Switch Language to Urdu

Name	Switch Language to Urdu
Actors	Farmer
Summary	Transitions application terminology formats into native Urdu representations.
Pre-Conditions	Preloaded typography assets exist inside the host bundle.
Post-Conditions	Total systematic transition of interface architecture text boundaries.
Special Requirements	Responsive Right-to-Left (RTL) matrix manipulation logic.
Basic Flow	
Actor Action	**System Response**
1. Chooses specific language toggle	2. Injects configuration flags across rendering states 3. Restarts navigational layout matrix aligned RTL in Urdu
Alternative Flow	
3a. In circumstances lacking explicitly translated keys, the rendering module conservatively inherits standard icon-driven or English fallbacks to prohibit blank rendering zones.	

4.7 Hardware and Software Requirements

4.7.1 Hardware Requirements

Deployment architecture demands an introductory-level low-budget Android smartphone possessing a baseline of 2GB operative RAM and 500MB storage capacity to isolate essential model artifacts and dataset caching. The system anticipates intermittent, yet unavoidable, network communication to synchronize historic elements. The backend service instances operate on server architectures characterized by high-concurrency capability, optimized compute resources, and rapid multi-threaded I/O environments.

4.7.2 Software Requirements

4.8 Graphical User Interface

Initial spatial constraints formulated over the foundational interactive screens intended for the Kisaan-Madad platform.

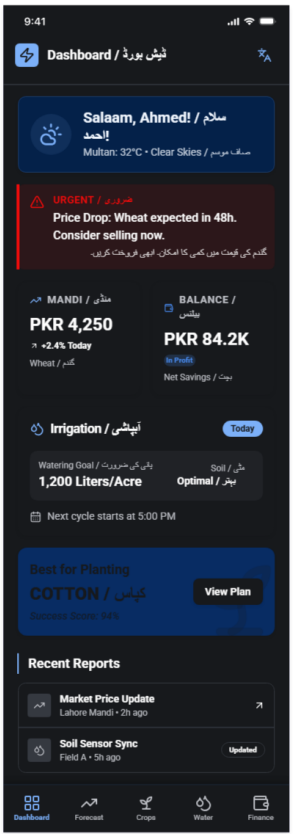

Figure 4.1: Wireframe: Home Dashboard

Table 4.8: Software Component and Technology Justification

Component	Technology Choice	Justification
Mobile Frontend	Jetpack Compose (Android)	Prioritises a native Android UI for the project's primary deployment target, enabling direct profiling and memory tuning on constrained handsets while avoiding the overhead of a fully shared UI runtime [4, 9, 5].
Shared Mobile Logic	Kotlin Multiplatform	Reuses stable non-UI components such as API clients, domain models, validation rules, and cache policies across Android and iOS without sacrificing native user interfaces [6].
Backend API	FastAPI (Python)	Selected for straightforward asynchronous endpoint development and tight integration with the Python-based machine learning workflow used by the forecasting module.
Price Forecasting Model	CNN-LSTM	Deep learning architectures are well suited to volatile agricultural time series, while the local market dynamics identified in Pakistan further justify a robust forecasting pipeline for mandi price prediction [19, 11].
Irrigation Module	FAO-56 + NASA POWER API	Integrating undisputed international standards [8] reinforced by verified reanalysis parameters [34].
Database	PostgreSQL	Provides reliable relational storage for farmer profiles, mandi price histories, irrigation records, and financial entries while supporting structured querying across linked entities.
ML Model Serving	TensorFlow Lite	Supports compact on-device inference for offline-capable components where model size, latency, and battery consumption must remain bounded on low-end smartphones.
Language Support	Jetpack Compose localization + Urdu font resources	Supports an Urdu-first interface aligned with the accessibility needs of farmers who rely on mobile phones for agricultural information but may face literacy and usability barriers in text-heavy systems [4, 16].

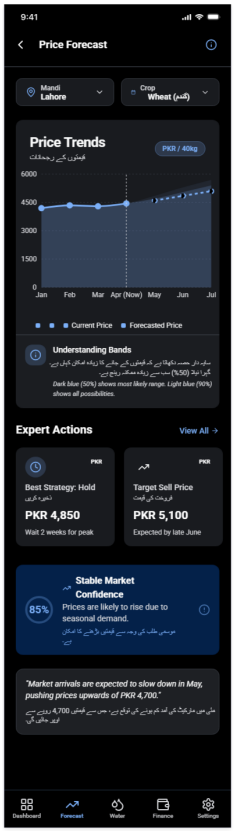

Figure 4.2: Wireframe: Price Forecast Screen

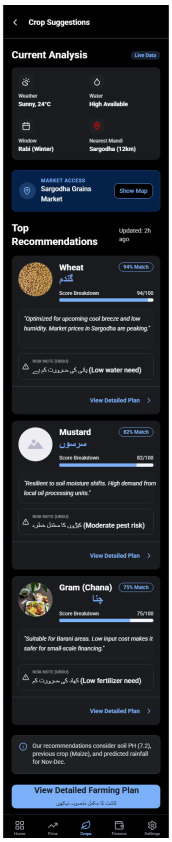

Figure 4.3: Wireframe: Crop Recommendation Screen

Figure 4.4: Wireframe: Irrigation Schedule Screen

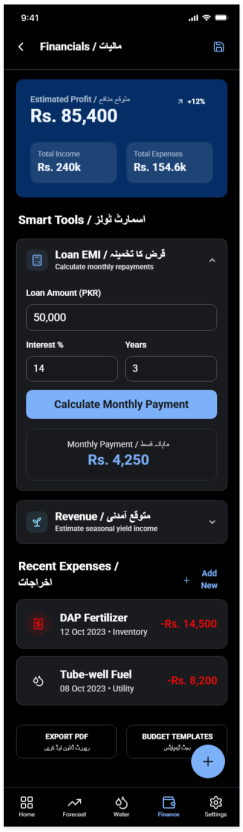

Figure 4.5: Wireframe: Financial Tools Screen

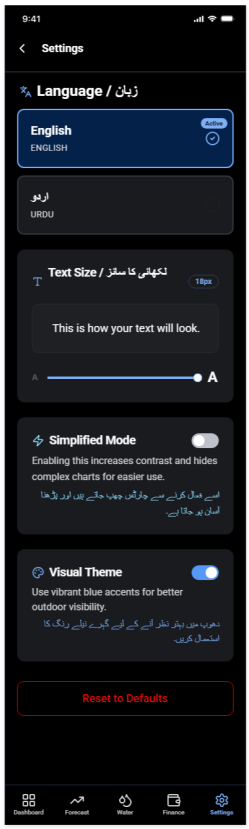

Figure 4.6: Wireframe: Language Settings Screen

4.9 Database Design

4.9.1 ER Diagram

The relational database topography fundamentally establishes **Farmer** as the core pivot entity directly corresponding to various secondary elements. A singular **Farmer** entity initiates a one-to-many link resolving downstream active **Crop** entities. Expanding horizontally, a registered **Crop** associates directly with daily chronological **MandiPrice** fluctuations while similarly mapping calculated **IrrigationSchedule** records. Abstract elements like **WeatherData** iterate asynchronously. Furthermore, granular financial operations spawn **LoanRecord** and **ExpenseRecord** parameters which fold linearly backward in many-to-one aggregation mapping to the original **Farmer** identifier.

Table 4.9: Data Dictionary: Farmer Entity

Attribute	Data Type	Constraints	Description
farmer_id	UUID	PRIMARY KEY	Distinct internal account token
phone_number	VARCHAR(15)	UNIQUE, NOT NULL	Formatted active MSISDN string
name	VARCHAR(100)	-	Alphabetical title sequence
location_lat	FLOAT	-	GPS geographic latitude parameter
location_long	FLOAT	-	GPS geographic longitude parameter
farm_size	DECIMAL(5,2)	> 0	Quantitative acreage domain
pref_language	VARCHAR(10)	DEFAULT 'UR'	Selected locale (UR/EN)

Table 4.10: Data Dictionary: MandiPrice Entity

Attribute	Data Type	Constraints	Description
price_id	SERIAL	PRIMARY KEY	Index sequential reference
crop_name	VARCHAR(50)	NOT NULL	Textual specific commodity
mandi_loc	VARCHAR(100)	NOT NULL	Active geographic trading zone
record_date	DATE	NOT NULL	Historical archival chronological stamp
min_price	DECIMAL(8,2)	-	Lowest formulated boundary mark
max_price	DECIMAL(8,2)	-	Highest formulated boundary mark
avg_price	DECIMAL(8,2)	NOT NULL	Arithmetic aggregated trading index

Table 4.11: Project Risk Configuration

Risk	Type	Likelihood	Impact	Mitigation Strategy
NASA POWER API downtime	Technical	Medium	High	Retain fallback methods mirroring the Hargreaves protocol executing independently over cached averages.
PAMRA data gaps	Data	High	Medium	Design processing mechanisms configured to resolve non-sequential blanks via structural interpolation matrices.
Low model accuracy on unseen crops	Technical	Low	High	Compartmentalize core forecasting implementations directly prioritizing solely wheat, cotton, and rice structures initially.
Low-end device performance friction	Technical	Medium	Medium	Constrain Compose recomposition surfaces, rely on lazy rendering and local caching, and shift heavier ML inference workloads to backend tiers whenever latency budgets allow.
Farmer digital literacy resistance	Business	High	High	Maintain rigid conformity enforcing comprehensive visual cues alongside ubiquitous Urdu integration natively.
Intermittent connectivity loops	Technical	High	High	Develop encompassing asynchronous cache engines isolating application operations from rigid dependency on connection states.
Data privacy vulnerabilities	Business	Low	Medium	Integrate complete data masking practices ignoring any extraneous data pooling beyond rigid minimal components.
Project scope creep	Business	Medium	Medium	Solidify and anchor subsequent structural iteration milestones strictly bound by approved specifications in Deliverable 2.

4.9.2 Data Dictionary

4.10 Risk Analysis

4.11 Conclusion

This chapter delineates the exhaustive software requirements parameterizing the robust functioning algorithms and constraints essential to KisaanMadad's implementation sequence. The detailed compilation clarifies functional expectations mapping into non-functional dependencies alongside preemptive risk assessment contingencies. Collectively encompassing hardware limits and interface accessibility metrics, the framework effectively predicates the impending system architectural design comprehensively addressed natively in Chapter 5.

Chapter 5 Proposed Approach and Methodology

This chapter details the foundational algorithmic approaches and computational methodologies driving the KisaanMadad integrated decision-support system. While the subsequent architectural design chapters delineate the structural flow of the application, this section mathematically and procedurally defines the core backend logic. Specifically, the framework is divided into four distinct analytical methodologies: the deep learning pipeline for probabilistic price forecasting, the multi-criteria scoring algorithm for crop recommendation, the standardized environmental equations for irrigation scheduling, and the deterministic rule-based calculators for financial planning. By formally defining these underlying models and their respective data ingestion pipelines, it is demonstrated how the system computationally addresses the localized agricultural uncertainties prevalent within Punjab, Pakistan.

5.1 Price Forecasting Methodology

The Price Forecasting module is designated to predict the stochastic nature of local mandi pricing for primary commodities (wheat, cotton, and rice) utilizing a probabilistic deep learning framework.

5.1.1 Data Acquisition and Preprocessing

The core methodology relies entirely on the extraction of daily wholesale pricing datasets sourced directly from the Punjab Agriculture Marketing Regulatory Authority (PAMRA) and the Agriculture Marketing Information Service (AMIS) spanning the temporal operational window of 2018 to 2023. These time-series datasets encapsulate extreme commodity volatility previously discussed. To prepare the sequences for sequential ingestion, missing entries are interpolated using localized linear imputation to prevent the destruction of temporal continuity. Subsequently, the univariate data is scaled utilizing Min-Max normalization, restricting the continuous values strictly between 0 and 1, ensuring computational stability during the gradient descent optimization phases.

5.1.2 CNN-LSTM Architecture

The predictive engine employs a hybrid Convolutional Neural Network and Long Short-Term Memory (CNN-LSTM) architecture. This approach, benchmarked against classical ARIMA and Prophet implementations [7], serves to capture distinct temporal profiles. The initial one-dimensional Convolutional (1D-CNN) layers isolate localized, short-term spatial pricing anomalies and drastic market shocks intrinsic to the mandi system. The extracted feature maps are subsequently transitioned into the LSTM layers, which are mathematically designed to prevent gradient vanishing and systematically learn extended temporal dependencies over the historical sequences. By processing these dual layers, the final

fully connected dense layer outputs a deterministic 30-day forward-looking prediction. Furthermore, utilizing dropout layers and randomized weight initializations, the module computes standard deviations formulating the upper and lower statistical confidence intervals necessary to present an actionable probabilistic range to the farmer.

5.2 Crop Recommendation Methodology

The Crop Recommendation module employs a Multi-Criteria Decision Making (MCDM) approach to synthesize diverse environmental and temporal factors, mapping optimal agricultural yields tailored to the specific constraints of the target farm acreage.

5.2.1 Input Parameters and NASA POWER Integration

The procedural algorithm initiates by requesting fundamental parameters from the farmer: land size, active district within Punjab, and intended planting season. Rather than exclusively relying on static historical templates, the module dynamically queries the NASA POWER API utilizing the district's geographical centroid. This query fetches localized meteorological parameters including cumulative precipitation averages, soil temperature thresholds, and solar radiation baselines.

5.2.2 Multi-Criteria Scoring Mechanism

The retrieved meteorological inputs are systematically evaluated against absolute growth thresholds delineated by standardized FAO crop calendars. The algorithm assigns weighted compatibility scores (S_c) for each candidate crop (c) based on four primary attributes: thermal suitability (W_T), precipitation alignment (W_P), market proximity and expected mandi demand (W_M), and temporal planting window accuracy (W_D). The final recommendation integer is computed via an additive aggregation:

$$S_c = \alpha W_T + \beta W_P + \gamma W_M + \delta W_D \tag{5.1}$$

where the coefficients α, β, γ, and δ represent the relative deterministic weightings isolated during the initial model tuning. Candidate crops are sorted in descending ordinal rank based on S_c, ensuring that the final output presented to the farmer explicitly reflects the most agronomically resilient and economically viable options for their highly specific microclimate and geographic restraints.

5.3 Irrigation Scheduling Methodology

Operating at the core of the system's resource management capabilities, the Irrigation Scheduling module is rigidly predicated upon the international FAO-56 Penman-Monteith standard [8].

5.3.1 Reference Evapotranspiration (ET_0)

The primary computational step requires determining the reference evapotranspiration (ET_0), signifying the environmental evaporative demand. The application pulls near-real-time absolute metrics—specifically maximum and minimum air temperature (T), solar radiation (R_s), atmospheric pressure, and approximate wind speed—directly from the NASA POWER API. The standard Penman-Monteith equation computes ET_0 in mm per day:

$$ET_0 = \frac{0.408\Delta(R_n - G) + \gamma \frac{900}{T+273} u_2 (e_s - e_a)}{\Delta + \gamma(1 + 0.34u_2)} \tag{5.2}$$

where Δ is the slope of the vapor pressure curve, R_n represents net radiation, G is soil heat flux, γ denotes the psychrometric constant, u_2 corresponds to wind speed, and $(e_s - e_a)$ defines the vapor pressure deficit.

5.3.2 Hargreaves-Samani Fallback

Recognizing the probability of API packet loss or incomplete NASA dataset rendering for wind speed and humidity variables, the methodology implements a graceful degradation mechanism via the Hargreaves-Samani equation. Relying exclusively on extraterrestrial radiation (R_a) and extreme localized atmospheric temperatures (T_{max}, T_{min}), the fallback estimates:

$$ET_0 = 0.0023 \cdot R_a \cdot (T_{mean} + 17.8) \cdot \sqrt{T_{max} - T_{min}} \tag{5.3}$$

This strategy ensures absolute stability for the continuous operation of the mobile device without compromising vital agricultural scheduling.

5.3.3 Crop Evapotranspiration (ET_c) and Scheduling

Subsequently, the system references predefined FAO statistical tables to lock the precise crop coefficient (K_c) identical to the active growth stage (initial, development, mid-season, or late season) of the farmer's designated commodity. Actual crop evapotranspiration (ET_c) is derived:

$$ET_c = K_c \times ET_0 \tag{5.4}$$

The aggregated weekly ET_c values translate directly into actionable volumetric fluid requirements, thereby allowing the module to push automated, mathematically precise watering schedules directly mitigating Pakistan's pressing hydrological volatility.

5.4 Financial Planning Methodology

Unlike the stochastic and environmental complexities governing the prior modules, the Financial Planning methodology employs strict, deterministic, and rule-based mathematical structures to provide complete economic clarity.

5.4.1 Loan Amortization Calculator

The financial architecture provides a standard cyclic loan calculator assessing the fiscal viability of banking advances. Accepting inputs for principal amount (P), annual interest rate (r), and total tenure in months (n), the engine calculates the equatable monthly installment (EMI):

$$EMI = P \times \left(\frac{r(1+r)^n}{(1+r)^n - 1} \right) \tag{5.5}$$

where r is mathematically reduced to the fractional monthly rate.

5.4.2 Revenue and Profit Projection

In parallel, the revenue engine cross-references the latest available PAMRA price indices against the farmer's estimated crop yield (Y) to formulate an expected gross revenue timeline. Additionally, it systematically tracks categorical expenses such as seeds, fertilizers, and arthi commission margins (E_{total}). Expected net profitability (N_p) is subsequently formulated:

$$N_p = (\text{Current Mandi Price} \times Y) - E_{total} \tag{5.6}$$

This deterministic tracking actively establishes the fiscal stability required for smallholders traditionally excluded from formal planning structures to optimize operational expenditures.

5.5 Conclusion

To conclude, the procedural approaches formulated to drive KisaanMadad are demonstrably anchored in a combination of predictive machine learning analysis, standardized environmental algorithms, and resolute mathematical logic. The application dynamically fuses localized economic variables gathered

from PAMRA and AMIS with real-time planetary metrics supplied by the NASA POWER API. By successfully encoding the probabilistic CNN-LSTM framework alongside the rigid FAO-56 Penman-Monteith methodology, the system transitions complex backend mechanics into actionable foresight. The established algorithms serve as the core intelligence foundation upon which the overarching system architecture and user interface, detailed in the subsequent chapter, are constructed.

Chapter 6 System Design and Architecture

This chapter details the high-level and low-level architectural design of the KisaanMadad integrated mobile decision-support system. The architectural representations illustrate the structural organization, sub-system interactions, and data flows essential to fulfilling the project's functional and non-functional requirements. By delineating the overarching system architecture alongside the specific domain models and design tactics, this chapter provides the blueprint guiding the technical implementation of the platform.

6.1 System Overview

The KisaanMadad platform is structured as a robust client-server architecture tailored for resource-constrained environments in Punjab, Pakistan. The frontend comprises a native Android application built with Jetpack Compose, complemented by Kotlin Multiplatform shared modules that encapsulate domain models, networking contracts, validation rules, and cache policies for eventual iPhone reuse. This Android-first choice reflects the practical deployment emphasis on low-cost handsets and the need for direct platform-level performance tuning, while still preserving a credible cross-platform path for iOS users [4, 9, 5, 6]. The mobile client interfaces asynchronously with a central backend API powered by FastAPI (Python), which serves as the primary orchestration layer. The backend manages local data persistence through a PostgreSQL relational database and securely brokers external requests to third-party data providers, notably the NASA POWER API for meteorological coordinates and PAMRA/AMIS for localized commodity pricing data. The overarching objective of this architecture is to minimize bandwidth consumption while ensuring high availability of agricultural predictive models on the edge.

6.2 Design Considerations

6.2.1 Assumptions and Dependencies

- **Connectivity Dependency:** The system assumes that while the farmer might operate in offline states, periodic connectivity is available to synchronize cached mandi datasets and synchronize the local FAO crop tables.
- **External API Stability:** Accurate irrigation modeling is strictly dependent on the continued availability and structural stability of the NASA POWER API endpoints.
- **Hardware Constraints:** The application design explicitly caters to low-memory Android devices common among smallholder farmers (i.e., devices with $<$ 2GB RAM), necessitating aggressive

memory management.

6.2.2 General Constraints

- **Limited Bandwidth:** Network payloads strictly consist of compressed JSON objects avoiding heavy graphical transmissions.
- **Multilingual Interfaces:** The software constraints require dynamic RTL (Right-to-Left) rendering capabilities to fully support the native Urdu typography cleanly.
- **Machine Learning Limitations:** Edge-based inferences using CNN-LSTM arrays must computationally complete within acceptable timeframes without draining the device's battery capacity.

6.2.3 Goals and Guidelines

- **Offline-First Functionality:** Vital calculations (like the Hargreaves fallback routing and financial calculators) must trigger unhindered without active server connections.
- **Scalability:** The foundational FastAPI layer dictates that new ML models or updated FAO guidelines can be integrated smoothly without forcibly halting operations.

6.3 System Architecture

The system architecture physically partitions responsibilities into a classic three-tier arrangement: Presentation Tier (Mobile App), Logic Tier (Backend Services/ML Processing), and Data Tier (Database and External APIs). This segregation prevents monolithic dependencies.

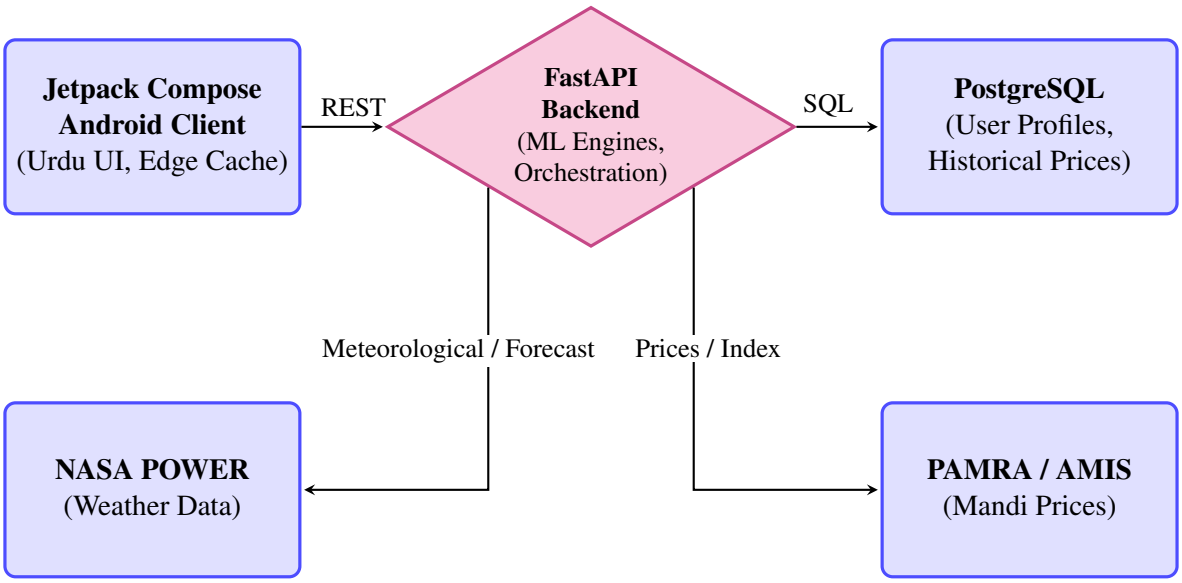

Figure 6.1: High-Level System Architecture of KisaanMadad.

The diagram conceptually portrays the data transmission cycle. The Jetpack Compose Android client securely requests data parameters via RESTful principles directly to the FastAPI backend, while Kotlin Multiplatform shared modules standardize request models, caching rules, and validation logic so that equivalent iPhone workflows can reuse the same non-UI core. The server autonomously evaluates whether the request implies retrieving stable Postgres elements or necessitates real-time network interactions scaling to NASA POWER for evapotranspiration mechanics.

6.3.1 Subsystem Architecture

The central backend comprises four modular logical nodes reflecting the product scope.

1. **Price Forecasting Subsystem:** Ingests historical sequences, normalizes numerical tensors, and executes the CNN-LSTM predictor to output structured JSON confidence intervals.
2. **Crop Recommendation Subsystem:** Houses the Multi-Criteria Decision Making matrix which cross-references incoming GPS latitudes alongside resident FAO calendars.
3. **Irrigation Scheduling Subsystem:** Hosts the Penman-Monteith algorithms and Hargreaves fallback models executing strictly on retrieved atmospheric conditions.
4. **Financial Planning Subsystem:** Operates as a purely deterministic mathematical orchestrator processing user-provided yield and loan parameters instantly.

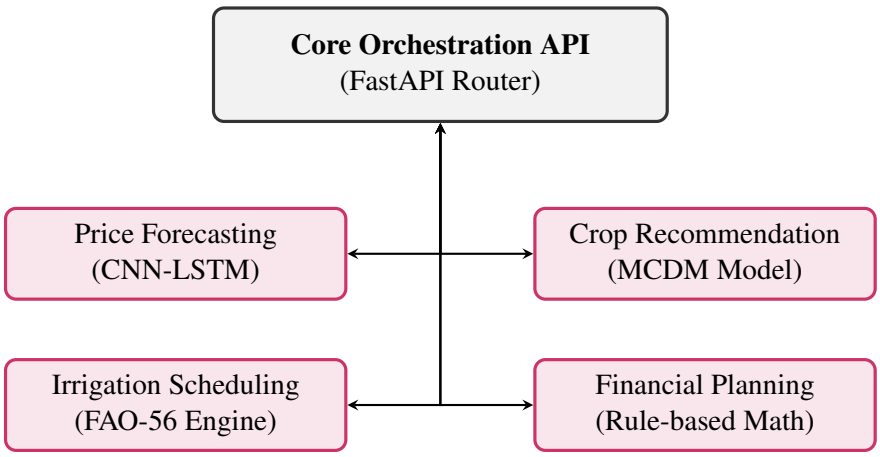

Figure 6.2: Subsystem modular architecture detailing feature pathways.

6.4 Architectural Strategies

6.4.1 Offline-First Mobile Caching

This strategy addresses rural network instability. The mobile client implements an internal SQLite storage layer. Whenever network latency spikes or connections drop, localized mathematical requests naturally revert to the recent cached matrices. The system transparently flags stale data to the user using discreet UI indicators.

6.4.2 Android-First Native UI with Shared Multiplatform Core

Because the primary field deployment concerns low-cost Android phones, the presentation layer remains native through Jetpack Compose rather than adopting a fully shared UI runtime. This decision permits direct use of Android profiling tools, fine-grained state management, and targeted memory optimization on constrained devices, thereby reducing performance friction on weaker handsets. To avoid duplicating stable business logic for iPhone users, networking, validation, serialization, and caching components are organized into Kotlin Multiplatform modules that can be consumed by a future native iOS client [9, 5, 6].

6.4.3 Graceful Degradation for Real-Time Models

For the FAO-56 irrigation protocol, real-time meteorological failures risk pausing vital decisions. The architectural strategy strictly delegates computation. If precise NASA solar radiation streams falter, the backend algorithm intercepts the error and routes the process block toward the temperature-limited Hargreaves-Samani equation, structurally guaranteeing an actionable numeric output.

6.5 Domain Model / Class Diagram

The structural representation explicitly models the relationships defining the software environment. It clarifies associative multiplicity existing between the primary entity *Farmer* and the underlying operational records.

6.6 Policies and Tactics

6.6.1 Asynchronous Executions

Given the heavy computational burden associated with CNN-LSTM processing sequences, local predictive tactics forbid synchronous blocking operations. The application UI systematically initiates progress

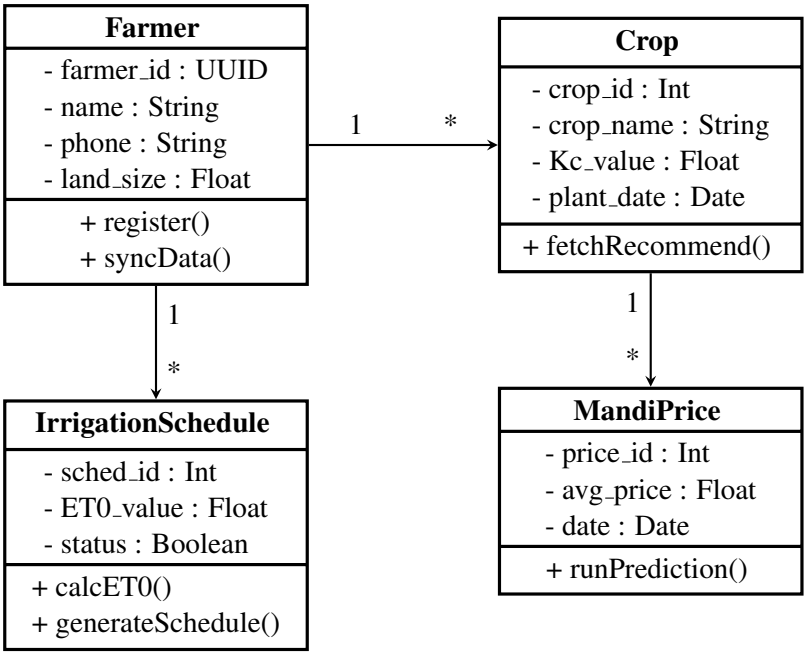

Figure 6.3: Domain Model / UML Class Diagram showing primary entity associations.

indicators corresponding to parallel thread executions, guaranteeing that visual presentation tiers never freeze during intense algorithm evaluations.

6.6.2 Strict Dependency Injection

The backend logic is rigorously encapsulated leveraging Dependency Injection (DI) paradigms. This structural tactic enforces that individual processing blocks (such as the API call to NASA POWER or database connection wrappers) exist completely decoupled. When deploying automated testing scripts during the QA phase, live dependencies are seamlessly replaced with localized mock arrays without restructuring underlying logic.

6.7 Conclusion

This chapter comprehensively detailed the overarching architecture managing KisaanMadad's execution matrices. The high-level deployment relies securely on asynchronous routing balancing a lightweight mobile client against sophisticated cloud operations. Deliberate structural fail-safes actively guarantee continuous decision-support capabilities despite severe environmental connectivity limits. The subsequent chapters will delineate the exact programmatic implementations and validation evaluations constructed upon these architectural blueprints.

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